

Request for Information (RFI) on Biologically-Derived Materials for Transient Propulsion Systems**Agency:** Office of the Director of National Intelligence**Office:** Intelligence Advanced Research Projects Activity**RFI Number:** IARPA-RFI-26-01

Synopsis

The Intelligence Advanced Research Projects Activity (IARPA) seeks information regarding the current state of the art (SOTA) in biologically-derived materials suitable for controlled transience in propulsion systems for unmanned aerial vehicles (UAVs). IARPA is interested in understanding how controlled material degradation can be extended beyond structural components to include turbines, engines, motors, and associated propulsion elements. Of particular interest are approaches that move beyond ultraviolet (UV)-initiated or water (H₂O)-initiated transience mechanisms and subsequently enable more diverse triggering modalities that are stable in a range of environmental conditions. This RFI is issued for planning purposes only and does not constitute a formal solicitation for proposals or suggest the procurement of any materials, components, or systems.

Background

The proliferation of UAV operations across military, intelligence, and civilian domains has raised growing concerns about the environmental footprint of aerospace materials, particularly in scenarios where systems may be lost. Conventional propulsion components, and their controls, fabricated from persistent materials—metal alloys, engineering plastics, and advanced composites—can remain in ecosystems for decades or centuries. DARPA's ICARUS (Inbound, Controlled, Air-Releasable, Unrecoverable Systems) program pioneered a solution for airframe structures by demonstrating transient materials that maintain operational performance before undergoing controlled degradation, thereby minimizing persistent environmental residue. The ICARUS approach employed primarily UV-triggered photopolymers that undergo programmable degradation of structural components, establishing proof-of-concept. However, extending these achievements beyond airframes to propulsion systems and their controls reveals critical limitations: propulsion components operate under extreme thermal, mechanical, and chemical stresses; contain a mixture of material types; must maintain structural integrity throughout dynamic mission profiles; require materials compatible with combustion environments or electromagnetic fields; and face significant challenges when UV initiation serves as the primary degradation trigger within enclosed or shielded propulsion assemblies where UV exposure may be limited or absent. Moreover, there may be operational parameters where natural UV exposure is present. Environmental conditions that naturally degrade materials—humidity, microbial activity, thermal cycling, oxidation—are often more readily available than UV exposure, suggesting

opportunities for bio-derived materials that leverage natural decomposition pathways to control degradation of an airborne system.

Biological systems offer compelling templates for addressing these challenges. Nature has evolved diverse materials that combine high-performance mechanical properties with controlled degradation pathways responsive to environmental triggers including enzymatic activity, pH, humidity, thermal cycling, and oxidative conditions. Examples span structural proteins (silk, keratin, collagen), polysaccharides (chitin, cellulose), fungal materials, bio-acrylics, and biocomposites that demonstrate remarkable strength-to-weight ratios while remaining inherently biodegradable under appropriate conditions.

Biologically derived materials including wood, paper, gums, cottons, and waxes still play a role in defense related materials, but recent advances in materials science, synthetic biology, and biomanufacturing have expanded the palette of engineered biological materials suitable for demanding applications. These include genetically modified structural proteins with tailored mechanical properties, bio-derived ceramics and composites for high-temperature applications, enzymatically-sensitive polymers with programmable degradation kinetics, and mycelium-based materials offering tunable porosity and mechanical performance. Yet significant gaps remain in translating these materials from laboratory demonstrations to operational propulsion systems that must reliably perform under extreme conditions before undergoing controlled transience.

Scope

This RFI specifically targets the unique challenges of propulsion systems and their controls, where materials must simultaneously satisfy competing requirements of high-performance operation and controlled degradation with limited degradation of UAV capability. IARPA seeks to understand potential pathways toward propulsion components fabricated from, coated with, or incorporating biologically-derived materials that enable:

1. **Operational performance** matching or approaching conventional propulsion materials during mission lifetime
2. **Controlled transience** initiated by environmental triggers beyond UV exposure
3. **Predictable degradation kinetics** allowing mission planning around component lifetime
4. **Minimal residue** following transience, leaving limited environmental signature
5. **Scalable manufacturing** compatible with rapid prototyping and production requirements

Propulsion Components of Interest

- Turbine/Engine Elements
 - Electric Motor Components
 - Auxiliary Systems
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Questions for Respondents

IARPA seeks improved understanding across the following areas. Responses addressing one, some, or all questions are encouraged:

Current SOTA Assessment

1. What biologically-derived materials have demonstrated mechanical, thermal, and chemical properties approaching those required for propulsion system components (e.g., turbine blades, combustion chambers, motor housings)?
2. What are the current performance limitations of bio-derived materials in high-temperature (>500°C), high-stress (>100 MPa), or chemically-aggressive environments relevant to propulsion systems?
3. What manufacturing approaches can produce bio-derived propulsion components with the dimensional tolerances, surface finishes, and quality control required for reliable operation?
4. What examples exist of bio-derived materials demonstrating controlled transience while maintaining operational performance? What degradation timescales have been achieved (hours, days, weeks)?

Material Systems and Architectures

5. What bio-derived material classes (proteins, polysaccharides, fungal materials, biocomposites, bio-acrylics, bio-ceramics) show greatest promise for propulsion applications? What are their operational temperature, strength, and durability limits?
6. Can hybrid architectures (bio-derived matrices with conventional reinforcements, or vice versa) provide transitional pathways toward fully-transient propulsion systems?
7. What surface treatments, coatings, or protective layers can enable bio-derived materials to withstand propulsion environments while maintaining transience capability?
8. How can hierarchical material structures inspired by biology (e.g., nacre, bone, wood) be engineered to provide both high performance and controlled degradation?

Transience Mechanisms and Triggers

9. What non-UV environmental triggers can initiate controlled degradation in bio-derived materials suitable for propulsion applications? How can trigger sensitivity be engineered?
10. How can degradation kinetics be tailored to match mission requirements (e.g., maintain integrity through mission duration, then degrade within hours of exposure to trigger)?
11. What approaches can provide fail-safe mechanisms ensuring transience even if primary triggers are not encountered? How can premature degradation be prevented?
12. Can enzymatic or microbial degradation be engineered to occur predictably across diverse environmental conditions? What biological agents enable reliable transience?

Integration and System Considerations

13. What propulsion system components represent the most tractable initial targets for bio-derived transient materials? Which pose the greatest technical challenges?
14. How do bio-derived propulsion materials affect overall system design (thermal management, lubricants, fuel compatibility, electromagnetic properties)?
15. What testing protocols and performance metrics are needed to validate bio-derived propulsion components for UAV applications? How can reliability be assessed?
16. What manufacturing scale-up challenges exist for bio-derived propulsion materials? What production timelines and costs are realistic?

Research Gaps and Opportunities

17. What fundamental research advances are needed to enable bio-derived transient propulsion systems? What are the highest-risk, highest-payoff research directions?
18. Where can synthetic biology, genetic engineering, or directed evolution accelerate development of materials tailored for transient propulsion applications?
19. What adjacent fields (biomineralization, structural biology, biomaterials, biomanufacturing) offer underutilized approaches applicable to this challenge?
20. What performance trade-offs are inherent in bio-derived transient materials, and how might mission concepts adapt to exploit their unique properties?

Preparation Instructions to Respondents

IARPA requests that respondents submit focused responses addressing one or more of the questions above. Submissions should include a brief technical description of relevant approaches, materials, or concepts; outline critical technical obstacles and potential pathways to address them; and comment on expected performance, limitations, and development timelines. Where appropriate, respondents may provide a non-proprietary rough order of magnitude (ROM) estimate of funding and resources required for research programs spanning one to four years. This announcement contains all information required to submit a response. No additional forms, kits, or other materials are needed.

IARPA welcomes responses from all capable and qualified sources including academia, industry, government laboratories, and federally funded research entities from within and outside the U.S. In response to submissions, IARPA may organize an invitation-only workshop for reviewing and discussing relevant research. Participants may be asked to make presentations based on their RFI submissions or associated research. Written submissions and workshop discussions may inform formulation of future U.S. Government research initiatives.

Formatting Requirements

Responses must meet the following specifications:

1. **Cover sheet** (one page) identifying the title, organization(s), and technical/administrative points of contact—including names, addresses, phone numbers, and email addresses of all co-authors—and clearly indicating association with IARPA-RFI-26-01.
 2. **Executive summary** (one-half page maximum) providing a substantive, focused overview;
 3. **Technical response** (maximum three pages) addressing one, some, or all questions posed above;
 4. **Detailed technical description** (maximum two pages) of technical challenges and potential research approaches relevant to biologically-derived transient propulsion materials;
 - A single quad-chart (PowerPoint slide or PDF) depicting the approach, key concepts, supportive illustrations, and potential impact may be included and does not count against the two-page limit;
 5. **References** (no page limit) with citations supporting significant claims or reports of success; and
 6. **Format specifications:** 12-point Times New Roman font, single-sided, single-spaced, 8.5 × 11-inch paper with 1-inch margins.
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Submission Instructions to Respondents

Responses to this RFI are due no later than **5:00 p.m. Eastern Time, May 15, 2026**. All submissions must be electronically submitted as PDF documents to **dni-iarpa-rfi-26-01@iarpa.gov**. Inquiries must be submitted to the same address. Do not send questions with proprietary content. No telephone inquiries will be accepted.

Disclaimers and Important Notes

This RFI is issued solely for information and planning purposes and does not constitute a solicitation or authorization to enter negotiations for a contract. IARPA is under no obligation to acknowledge receipt of information or provide feedback to respondents. Responses are not offers and cannot be accepted by the Government to form binding contracts. Respondents are solely responsible for all expenses associated with responding. IARPA will not provide reimbursement for costs incurred. Respondents must ensure submitted materials are approved for public release by the information owner. The Government does not intend to award contracts based on this RFI or pay for information solicited, nor is it obligated to issue solicitations based on responses received.

No proprietary and no classified concepts or information shall be included in submittals. Should respondents wish to submit classified information, prior coordination must be made with the IARPA Program Security Officer via email to the Primary Point of Contact. Input on technical aspects may be solicited by IARPA from non-Government consultants/experts bound by appropriate non-disclosure requirements.

Contracting Office Address

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Primary Point of Contact

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